

Bucket Brigade: A Parametric Cooperative-Competitive Benchmark with Computable Nash Equilibrium Structure

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Abstract

Canonical cooperative-MARL benchmarks (Overcooked, Hanabi, SMAC, Melting Pot, MAgent, PettingZoo) trade equilibrium transparency for environment richness: their state spaces are large enough that no Nash equilibrium is computable, so when a new algorithm beats PPO on Overcooked the reader cannot tell whether the algorithm has converged to the “right” joint policy or merely to a Pareto-different one. *Bucket Brigade* inverts this trade. It is a finite parametric cooperative-competitive stochastic game on a ring of houses whose per-agent action space at the minimal parameterization is 8, whose state space is 2304 states, and whose three load-bearing scalars (β, κ, c) control equilibrium structure continuously across cells. We give the environment specification, derive closed-form boundary inequalities for the three Nash-equilibrium regimes (symmetric all-Work, asymmetric 1-Worker, no-pure-NE collapse) via a single-house mean-field reduction, and report empirical agreement between predicted and PPO-realised verdicts on a seven-cell preview of the phase diagram. The analytical reduction correctly predicts the qualitative phase order (collapse $\rightarrow$ symmetric $\rightarrow$ mixed $\rightarrow$ asymmetric) and β -independence at $c=0.5$, and the trainability sweep recovers the same ordering under a per-cell NE-anchored metric: `gap_closed_ne` is largest on the symmetric-only row (0.326), positive but smaller on the asymmetric-only row (0.106), and near-zero on the no-pure-NE row. The quantitative κ -thresholds of the analytical reduction disagree with the equilibrium solver by 3–10 $\times$; we own this gap and attribute it to ring-locality, per-agent ownership rewards, and the heuristic-strategy approximation. We position Bucket Brigade as a *methodological complement* to the surveyed benchmarks, not a replacement: it is the only published parametric MARL game where the question “did the algorithm converge to the right equilibrium?” admits a ground-truth answer.

1 Introduction

Multi-agent reinforcement learning evaluates new algorithms against benchmarks whose Nash-equilibrium structure is not characterizable in closed form. Overcooked-AI (Carroll et al., 2019), DeepMind’s Melting Pot (Leibo et al., 2021), the Hanabi Challenge (Bard et al., 2020), SMAC and SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019), MAgent (Zheng et al., 2018), and the PettingZoo MPE suite (Terry et al., 2021) between them cover small cooperative coordination, partial observability, scripted-opponent micromanagement, and population-scale competition. None of them ship with a method that produces the equilibrium strategy profile of the as-designed environment. When PPO fails to learn a coordinated policy on Overcooked, the community is left to debate whether the failure is the algorithm or the absence of a clean attractor for self-play; neither hypothesis can be rejected because no oracle exists for “what should the learned policy look like.”

We introduce *Bucket Brigade*, a finite parametric cooperative-competitive stochastic game designed to invert this trade. At its minimal parameterization the state space is 2304 states

and the per-agent action space is 8, so the one-shot stage game is enumerable and standard equilibrium-computation tools actually run. Across a smooth three-scalar parameter family (β, κ, c) the equilibrium structure transitions from symmetric all-Work to asymmetric 1-Worker (the textbook free-rider problem) to no-pure-NE collapse. The ground-truth target for each parameter cell is known, so a failed PPO run is attributable to the algorithm rather than to ambiguity about what convergence would mean. The pitch is exactly a *methodological complement* to the six canonical benchmarks above: those exist for studying open-ended cooperation, generalization, and scale; Bucket Brigade exists for studying equilibrium-convergence quality on a controlled grid.

Contributions. (1) A self-contained formal specification of the Bucket Brigade game (§2), recovering Volunteer’s Dilemma, the N -player Public Goods game, and Stag Hunt as named limits. (2) Closed-form boundary inequalities (§3) for the three NE regimes via a single-house mean-field reduction, together with an explicit accounting of where the closed-form thresholds disagree with empirics. (3) A protocol, a per-cell NE-anchored evaluation metric, and the first empirical results (§4) testing PPO trainability against the NE phase diagram on the 7-cell preview. The PPO ordering recovered by the per-cell metric matches the analytical prediction $\text{symmetric} > \text{asymmetric} > \text{collapse}$. (4) An honest positioning (§5, §6) against the canonical benchmarks: Bucket Brigade is small and artificial by design, and is not a substitute for any of them. Throughout, we treat the analytical NE characterization as ground truth and the PPO trainability sweep as a falsification test, in that order.

2 The Bucket Brigade environment

Players and topology. A finite set of agents $\mathcal{I}=\{1, \dots, N\}$ inhabits a cycle graph $\mathcal{H}=\mathbb{Z}/H\mathbb{Z}$ of H houses with neighbour edges. Each house has status $h_h \in \{\text{SAFE}, \text{BURNING}, \text{RUINED}\}$; RUINED is absorbing. The default “10-house” scenario fixes $H=10, N=4, T_{\min}=12$; the minimal scenario fixes $H=2, N=4$.

Action and observation. Each agent i chooses $a_{t,i}=(a^{\text{house}}, a^{\text{mode}}, a^{\text{sig}}) \in \mathcal{H} \times \{0, 1\} \times \{0, 1\}$: a target house, a WORK/REST bit, and a one-bit broadcast signal observable to all agents on the next night. Per-agent action cardinality is $4H$ (8 at $H=2$; 40 at $H=10$). Joint-action cardinality at the minimal parameterization is $8^4=4096$. Observation is symmetric perfect monitoring with one-step signal delay: every agent observes the full state plus the previous joint $(\ell_{t-1}, \alpha_{t-1}, \zeta_{t-1})$. There is no partial observability of houses or agent positions; the signal channel is non-binding cheap talk.

Transition dynamics. A single night consists of seven sequential phases applied in a fixed, load-bearing order: (1) signal write, (2) location/mode write, (3) extinguish, (4) burn-out, (5) spread, (6) spontaneous ignition, (7) reward and termination check. In the extinguish phase, if $w_h(a)$ agents WORK at a BURNING house, the house transitions to SAFE with probability $1 - (1 - \kappa)^{w_h(a)}$ (independent-workers model). Every BURNING house that survives the extinguish phase transitions deterministically to RUINED in the burn-out phase. The spread phase then ignites SAFE neighbours of BURNING houses with probability β per edge; under the canonical order, the burn-out phase has already cleared all BURNING sources, so spread is inert in the within-night dynamics and contagion enters only via the spontaneous-ignition phase 6 at rate ρ . (A contagion-active variant that swaps phases 4 and 5 is available; the phase-diagram analysis below uses the canonical order.)

Reward structure. The per-agent reward decomposes into three independently-parameterised terms: a private work/rest cost ($-c$ for WORK, $+c_{\text{rest}}=0.5$ for REST; the free-rider gradient is the $c+c_{\text{rest}}$ asymmetry), a per-house ownership term (r_{own} paid on save, p_{own} recurring on every

RUINED night), and a team welfare term proportional to the fraction of SAFE and RUINED houses. Ownership is assigned round-robin ($\text{owner}(h)=h \bmod N$). The asymmetric reward variants in the calibrated family let $r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}}$ be set per-agent, breaking exchangeability.

Relationship to canonical templates. Bucket Brigade recovers, as named limits or restrictions, several textbook templates: the N -player Volunteer’s Dilemma at $H=1$; the N -player Public Goods game when ownership weights collapse to uniform; Stag Hunt in the $\kappa \rightarrow 1$ limit; and a free-rider problem at the parameter cells where the only NE is asymmetric (the “rest-trap”). Formal proofs and a complete notation table are deferred to the env-spec memo (Walters, 2026a); §2 above is the minimal contract a reader needs to follow §3.

3 Equilibrium structure: closed-form boundaries

The central contribution of this paper is a closed-form characterisation of which Nash-equilibrium structure appears as a function of (β, κ, c) in the 4-agent game. We derive it via a single-house mean-field reduction that collapses the $H=10$ ring to a representative single house with all $N=4$ agents present, absorbs multi-step contagion and ignition dynamics into a per-night survival coefficient, and reads off the NE candidates as algebraic inequalities.

Reduction. Fix the `minimal_specialization` reward tuple $W=(R_{\text{team}}, P_{\text{team}}, r_{\text{own}}, r_{\text{other}}, p_{\text{own}}, p_{\text{other}})=(10, 10, 5, 5, 10, 10)$ and $\rho=0.02, T_{\min}=12$. With k workers at the house on a given night, the per-night ruin probability for an initially-SAFE house is $q(k)=\rho(1-\kappa)^k$ (under the canonical phase order, where β does not enter the leading-order single-night payoff). Integrating the lost per-house reward stream over $T_{\min}=12$ nights and collecting terms yields

$$\mathbb{E}[\text{per-house reward loss} \mid k] = A \cdot \rho \cdot (1 - \kappa)^k,$$

with $A=r_{\text{own}}T_{\min} + p_{\text{own}}T_{\min} + P_{\text{team}}T_{\min}/H=1812$ and survival coefficient $\tilde{A}:=A\rho=36.24$. The marginal benefit of adding the k -th worker is $\Delta(k)=\tilde{A} \cdot \kappa \cdot (1 - \kappa)^k$.

The three NE candidates. Writing $c_{\text{gap}}:=c + c_{\text{rest}}$ (so $c_{\text{gap}}=1.0$ when the driver sets $c=0.5$), the deviation comparisons against the per-night stage game under stationary opponent policies yield three inequalities.

The *symmetric all-Work* NE exists iff a unilateral deviation from $k=4$ to $k_{-i}=3$ does not pay:

$$\tilde{A} \cdot \kappa \cdot (1 - \kappa)^3 \geq c_{\text{gap}}. \quad (\text{S})$$

The *asymmetric 1-Worker* NE exists iff the lone worker prefers WORK ($k_{-i}=0$ deviation) and each free-rider prefers REST ($k_{-i}=1$ deviation):

$$\tilde{A} \cdot \kappa \geq c_{\text{gap}} \quad \text{AND} \quad \tilde{A} \cdot \kappa \cdot (1 - \kappa) < c_{\text{gap}}. \quad (\text{A})$$

The *no-pure-NE collapse* regime holds whenever even a lone worker cannot recover their cost, so neither (S) nor (A) can be satisfied:

$$\tilde{A} \cdot \kappa < c_{\text{gap}}. \quad (\text{C})$$

Predicted thresholds. Substituting $\tilde{A}=36.24, c_{\text{gap}}=1.0$ into (S)–(C): the collapse boundary (C) sits at $\kappa \approx 0.028$; the symmetric NE exists on $\kappa \in [0.030, 0.65]$ via (S); the asymmetric NE exists for $\kappa \gtrsim 0.972$ via (A). The qualitative phase order $\kappa\text{-small} \rightarrow \kappa\text{-large}$ is **collapse** $\rightarrow$ **symmetric_only** $\rightarrow$ **mixed** $\rightarrow$ **asymmetric_only**, and the derivation is structurally β -independent under the canonical phase order.

Predicted vs. observed thresholds. A 7-cell preview of the empirical phase diagram (computed by a heterogeneous Double-Oracle solver over continuous heuristic-parameter strategies; $c=0.5, \rho=0.02$; Figure 1) recovers the qualitative phase order and the β -independence at $c=0.5$ (3/3 κ -rows show identical verdicts across β). However, the *quantitative* κ -thresholds disagree with the predictions: the empirically-observed collapse boundary sits near $\kappa \approx 0.3$ (3–10 $\times$ above the predicted 0.028), and the empirically-observed asymmetric-NE onset sits near $\kappa \approx 0.7$ (below the predicted 0.972). Only 2 of 7 cells (both at $\kappa=0.5$) achieve exact verdict agreement.

We *own* this gap. The mean-field reduction has three identifiable systematic biases. (i) *Ring locality*: the derivation puts all four agents at one house, but empirically four agents distribute across ten houses, so the effective per-house defence is $1-(1-\kappa)^{0.4}$ rather than $1-(1-\kappa)^4$ —roughly a 4 $\times$ weaker defence at $\kappa=0.5$, which shifts the collapse boundary substantially upward into the empirically-observed range $\kappa \in [0.1, 0.3]$. (ii) *Per-agent ownership rewards*: the analytical A coefficient lumps r_{own} and p_{own} into the survival term, but the per-agent ownership reward only fires on the agent’s own house, scaling the off-own-house contribution down by a factor of roughly $1/H$. (iii) *Heuristic strategy space*: the empirical solver’s “free-riders” are **Firefighter** agents at `work.tendency=0.9`, not literal REST agents, so the effective k_{-i} in the lone-worker’s deviation comparison is closer to 3.7 than to 1. None of these corrections are derivable in closed form without losing the algebraic tractability that motivated the reduction; the natural extensions are named in §6.

Headline finding. The mean-field reduction correctly predicts *three* structural properties of the empirical phase diagram—the qualitative phase order, β -independence under the canonical phase order, and the existence of a no-pure-NE collapse regime at small κ —none of which would be derivable without the closed-form derivation. The quantitative κ -thresholds are off by 3–10 $\times$; we report this disagreement as the paper’s claim, not as a weakness to hide. The qualitative phase order furthermore transfers to the PPO trainability sweep under a per-cell NE-anchored metric (§4). The per-cell predicted-vs-empirical table and the full bias accounting appear in (Walters, 2026b).

4 Trainability: PPO against the NE phase diagram

Protocol. We trained independent PPO runs on each of the 7 phase-diagram cells from Figure 1 using the in-repo `JointPPOTrainer` on the `minimal_specialization` scenario, with 50 training iterations of 2048 rollout steps and 4 random seeds per cell ($N=28$ runs). Hyper-parameters are pinned in `experiments/p3_specialization/`; the sweep ran on a single 16-core host in approximately two hours. The PPO implementation follows the standard recipe of Schulman et al. (2017); the multi-agent joint-policy formulation follows the MAPPO-style centralised-critic setup of Yu et al. (2022). The cells from (Walters, 2026b) contribute the NE verdict for each (β, κ) combination tested.

Per-cell NE-anchored metric. The original v1 sweep scored every cell against a single canonical baseline (the `Random` and `SpecialistPolicy` returns measured at one calibration cell, $\beta=0.25, \kappa=0.5$). That metric was misleading across cells because the random-policy return varies substantially with κ (a low- κ cell is intrinsically harder for any policy, so a small absolute gap against the canonical-cell baseline understates relative PPO success). We replaced it with `gap_closed_ne`, the per-cell gap-closure ratio between `Random` and the NE *policy profile* recovered for that cell by the heterogeneous Double-Oracle solver (typically 1 $\times$ **Hero** + 3 $\times$ **Firefighter** on asymmetric-only cells, and 4 $\times$ **Hero** on symmetric-only cells). The per-cell baselines are tabulated in `experiments/nash/phase_diagram/per_cell_baselines.json` and computed by `bucket_brigade/baselines/pe`

the recipe is the same Random/SpecialistPolicy methodology of (Walters, 2026b), applied per cell. The methodological lesson is independent of this paper’s specific claims: single-cell baselines mislead cross-cell PPO comparison whenever the random-policy return is itself a function of the cell’s parameters, and per-cell calibration is the necessary correction. We report this as a separate contribution alongside the data themselves.

Results. Figure 2 reports the per-cell `gap_closed_ne` (mean $\pm$ std over 4 seeds). The ordering across NE classes is `symmetric_only` (0.326) > `asymmetric_only` (0.106) > `no_convergence` (-0.033 ; here the metric falls back to `gap_closed_homogeneous` because no NE policy exists), in exact agreement with the analytical prediction of §3. The two $\kappa=0.5$ cells— $\beta=0.5$ and $\beta=0.9$ —differ by 0.017, well within seed variance, which is consistent with the β -independence prediction of (S)–(C) at the PPO level: the per-cell std (≈ 0.33 over 4 seeds) exceeds the cell-to-cell difference by an order of magnitude, so the PPO data does not reject β -independence but also does not independently establish it. The equilibrium-solver verdicts (3/3 κ -rows identical across β ; §3) carry the solver-level evidence. PPO recovers substantial value on symmetric-only cells, partial value on asymmetric-only cells, and approximately none on no-convergence cells, consistent with the hypothesis that PPO’s policy-iteration loop converges reliably when a clean symmetric attractor exists, struggles when the only NE requires symmetry-breaking, and fails when no pure-strategy NE is available to converge to.

A methodological observation we report as a separate contribution: under the original single-cell baseline of the v1 draft, the ordering inverted to `asymmetric_only` (0.262) > `symmetric_only` (0.091) > `no_convergence` (-0.033) which would have falsified the §3 prediction. The inversion is an artefact of the metric, not the algorithm: the asymmetric-only cells at $\kappa=0.9$ have a much lower random-policy return than the canonical baseline cell, so PPO’s *absolute* improvement looked large despite recovering a smaller fraction of the available NE value. Per-cell baselines resolve the ambiguity and the corrected ordering matches the analytical prediction.

Caveats. The PPO sweep covers 7 cells, not the 75 cells of the planned full grid ($\beta \in \{0.1, 0.25, \dots, 0.9\} \times \kappa \in \{0.1, \dots, 0.9\} \times c \in \{0.5\}$); within-class statistics are therefore small ($n=2$ cells for the symmetric row, $n=2$ for asymmetric, $n=3$ for collapse). The PPO budget is also modest by recent MARL standards (50 iterations $\times$ 2048 rollout steps); a longer-budget sweep is in scope for the camera-ready and may shift absolute `gap_closed_ne` magnitudes without changing the ordering. The `gap_closed_homogeneous` metric (per-cell Random $\rightarrow$ SpecialistPolicy $\times 4$) reports quantitatively smaller cross-cell separation than the NE-anchored metric, but yields the same qualitative ordering on the 7-cell preview, so the load-bearing claim is robust to the metric choice.

5 Related work

The MARL community has converged on a handful of canonical benchmarks that trade equilibrium transparency for environment richness. Overcooked-AI (Carroll et al., 2019) is a fully cooperative coordination game on small grid layouts: every Pareto-optimal joint policy is a Nash equilibrium and “the” target is therefore not unique. Melting Pot (Leibo et al., 2021) exposes $88 \times 88 \times 3$ RGB observations across more than 50 substrates; its design explicitly does not solve for equilibrium structure on individual substrates. The Hanabi Challenge (Bard et al., 2020) has near-optimal hat-guessing strategies known at 2 players, but the 4–5 player game is open and no algorithm reaches the equilibrium from learning. SMAC and SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019) are cooperative micromanagement against a scripted opponent and “NE” is not a design target. MAgent (Zheng et al., 2018) hosts thousand-agent battles on grids and no published equilibrium

Benchmark	Year	Agents	Per-agent $ \mathcal{A} $	NE characterisable?	Coop
Overcooked-AI (Carroll et al., 2019)	2019	2	6	No (any Pareto-opt.)	coop
Melting Pot (Leibo et al., 2021)	2021	2–16	8	No (intractable)	mixe
Hanabi (Bard et al., 2020)	2020	2–5	≤ 20	Partial (2P only)	coop
SMAC/SMACv2 (Ellis et al., 2023; Samvelyan et al., 2019)	2019/23	2–27	$6+n_e$	No	vs. s
MAgent (Zheng et al., 2018)	2018	$64\text{--}10^3$	21	No (population)	mixe
PettingZoo MPE (Terry et al., 2021)	2021	2–6	3–5	No	per-s
Bucket Brigade (this work)	2026	4 (def.)	8 / 40	Yes (closed-form)	mixe

Table 1: Bucket Brigade against the six most-cited canonical MARL benchmarks. The right-most column “NE characterisable?” is the only axis on which Bucket Brigade dominates; on every other axis the surveyed benchmarks are richer, more cited, or larger in scope. This is the entire pitch: Bucket Brigade is a *methodological complement*, not a replacement. Per-agent action sizes for Bucket Brigade are 8 at the $H=2$ minimal scenario and 40 at the $H=10$ default.

analysis exists at scale. PettingZoo’s MPE suite (Terry et al., 2021) is the canonical small multi-agent baseline with no equilibrium characterisation across scenarios.

Table 1 compares Bucket Brigade against these on six axes.

The honest comparison is that Bucket Brigade dominates on exactly one column—NE characterisability—and loses on every other (citation footprint, agent count, environment richness, generality). The paper’s job is to argue that the NE-characterisability column matters enough to be worth introducing a new benchmark despite the trade. Bucket Brigade does not scale to large populations, does not test “emergent cooperation” in any general sense, and is not a general-purpose MARL benchmark. The niche it fills is the specific question of *equilibrium-convergence quality*.

6 Discussion and limitations

Bucket Brigade is intentionally small and artificial. The state space is 2304 at the minimal scenario and $\sim 10^{10}$ at the default 10-house scenario, the geometry is a 1-D ring, and the environment ships from a single research repository with no community adoption. We acknowledge each point head-on and defend it as the cost of the equilibrium-transparency property: (a) *Size is the cost of NE transparency.* A richer game with realistic observations, partial observability across many channels, and complex spatial dynamics would lose the property that motivated the work; the state-space cardinality is what makes the analytical reduction in §3 algebraically tractable. Trading richness for tractability is the deliberate design choice. (b) *A custom benchmark is acceptable given the methodological niche.* The question we want to answer—“did this MARL algorithm converge to the analytically-characterised NE?”—cannot be asked of any of the six canonical benchmarks in Table 1. A benchmark designed for this question is therefore a methodological contribution distinct from the contributions those benchmarks make, even if it does not inherit their scale or citation impact. (c) *Bucket Brigade is not a substitute for any of the six.* It does not replace Overcooked for human-AI coordination, Melting Pot for generalisation across substrates, Hanabi for partial-observability cooperation, SMAC for micromanagement against scripted opponents, MAgent for population-scale phenomena, or PettingZoo for the small-multi-agent baseline role. A researcher choosing a benchmark for any of those purposes should not pick Bucket Brigade.

Threats to the analytical contribution. The closed-form thresholds disagree with empirics by $3\text{--}10\times$ (§3). The three identified systematic biases—ring locality, per-agent ownership rewards, and the heuristic-strategy approximation—are not separated by the 7-cell preview; rigorous separation requires the full 75-cell grid (in progress) and at least one ρ -sweep to falsify the β -independence

prediction outside the small- $\beta\rho$ regime. The mean-field reduction is the load-bearing simplification that buys tractability; tightening it without losing tractability (e.g. via a ring-Markov refinement of the survival coefficient A) is the natural extension.

Threats to the empirical contribution. The PPO ordering reported in §4 matches the analytical prediction on the 7-cell preview, but the per-class sample size is small ($n \leq 3$ cells per NE class) and the budget per run is modest. The full-grid sweep at a longer budget—scheduled for the camera-ready—may shift the absolute `gap_closed_ne` magnitudes without disturbing the ordering, or may surface within-class heterogeneity that the preview cannot resolve. The methodological result—that per-cell baselines are required for cross-cell PPO comparison in cells where the random-policy return varies—stands independently of the eventual full-grid outcome.

Reproducibility. The environment is shipped as a pip-installable Python package (`pip install bucket-brigade`); the figure-by-figure reproduction recipe lives in the in-repo `docs/PAPER_RESULTS.md`. The per-cell baselines used by the §4 metric are produced by `bucket-brigade/baselines/per_cell.py` and serialised at `experiments/nash/phase_diagram/per_cell_baselines.json`; both ship with the repository so the recalibrated `gap_closed_ne` column of Figure 2 can be re-derived from the raw PPO-sweep checkpoints. Frozen baseline checkpoints (the analytical Hero, Firefighter, and FreeRider agents used to anchor the phase-diagram solver and the per-cell PPO metric) will be hosted on the HuggingFace Hub; the publication pathway for the baselines is in progress and not yet complete at the time of this draft. The source repository is at <https://github.com/rjwalters/bucket-brigade>.

Conclusion

We presented Bucket Brigade, a parametric cooperative-competitive MARL benchmark whose Nash-equilibrium structure is characterisable in closed form across a smooth three-scalar parameter family. A single-house mean-field reduction yields algebraic boundary inequalities (S)–(C) that correctly predict the qualitative phase order and the β -independence under the canonical phase order, while disagreeing with empirics on the quantitative κ -thresholds by 3–10 $\times$ —a disagreement we own and attribute to three named systematic biases. The 7-cell PPO trainability sweep, evaluated under a per-cell NE-anchored metric, recovers the predicted ordering `symmetric > asymmetric > collapse`, and the per-cell baseline correction itself is a methodological contribution applicable to cross-cell PPO comparison more broadly. Bucket Brigade is positioned as a methodological complement to the canonical MARL benchmarks for the specific question of equilibrium-convergence quality, not as a replacement for any of them.

References

- Nolan Bard, Jakob N. Foerster, Sarath Chandar, Neil Burch, Marc Lanctot, H. Francis Song, Emilio Parisotto, Vincent Dumoulin, Subhodeep Moitra, Edward Hughes, Iain Dunning, Shibl Mourad, Hugo Larochelle, Marc G. Bellemare, and Michael Bowling. The Hanabi challenge: A new frontier for AI research. *Artificial Intelligence*, 280, 2020.
- Micah Carroll, Rohin Shah, Mark K. Ho, Thomas L. Griffiths, Sanjit A. Seshia, Pieter Abbeel, and Anca Dragan. On the utility of learning about humans for human-AI coordination. In *Advances in Neural Information Processing Systems*, 2019.

- Benjamin Ellis, Jonathan Cook, Skander Moalla, Mikayel Samvelyan, Mingfei Sun, Anuj Mahajan, Jakob Foerster, and Shimon Whiteson. SMACv2: An improved benchmark for cooperative multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems*, 2023.
- Joel Z. Leibo, Edgar A. Dueñez-Guzmán, Alexander S. Vezhnevets, John P. Agapiou, Peter Sunehag, Raphael Koster, Jayd Matyas, Charles Beattie, Igor Mordatch, and Thore Graepel. Scalable evaluation of multi-agent reinforcement learning with Melting Pot. In *International Conference on Machine Learning*, 2021.
- Mikayel Samvelyan, Tabish Rashid, Christian Schroeder de Witt, Gregory Farquhar, Nantas Nardelli, Tim G. J. Rudner, Chia-Man Hung, Philip H. S. Torr, Jakob Foerster, and Shimon Whiteson. The StarCraft multi-agent challenge. In *Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems*, 2019.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.
- J. K. Terry, Benjamin Black, Nathaniel Grammel, Mario Jayakumar, Ananth Hari, Ryan Sullivan, Luis S. Santos, Clemens Dieffendahl, Caroline Horsch, Rodrigo Perez-Vicente, Niall L. Williams, Yashas Lokesh, and Praveen Ravi. PettingZoo: Gym for multi-agent reinforcement learning. In *Advances in Neural Information Processing Systems*, 2021.
- Robb Walters. Bucket brigade — formal environment specification (internal memo), 2026a. Internal anvil memo, `paper/anvil_memo.env_spec.1/`; to be released as arXiv preprint before submission.
- Robb Walters. Analytical NE characterization for 4-agent Bucket Brigade (internal memo), 2026b. Internal anvil memo, `paper/anvil_memo.ne_structure.1/`; to be released as arXiv preprint before submission.
- Chao Yu, Akash Velu, Eugene Vinitzky, Jiaxuan Gao, Yu Wang, Alexandre Bayen, and Yi Wu. The surprising effectiveness of PPO in cooperative multi-agent games. In *Advances in Neural Information Processing Systems*, 2022.
- Lianmin Zheng, Jiacheng Yang, Han Cai, Ming Zhou, Weinan Zhang, Jun Wang, and Yong Yu. MAgent: A many-agent reinforcement learning platform for artificial collective intelligence. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2018.

Heterogeneous Nash Phase Diagram — preview/primary

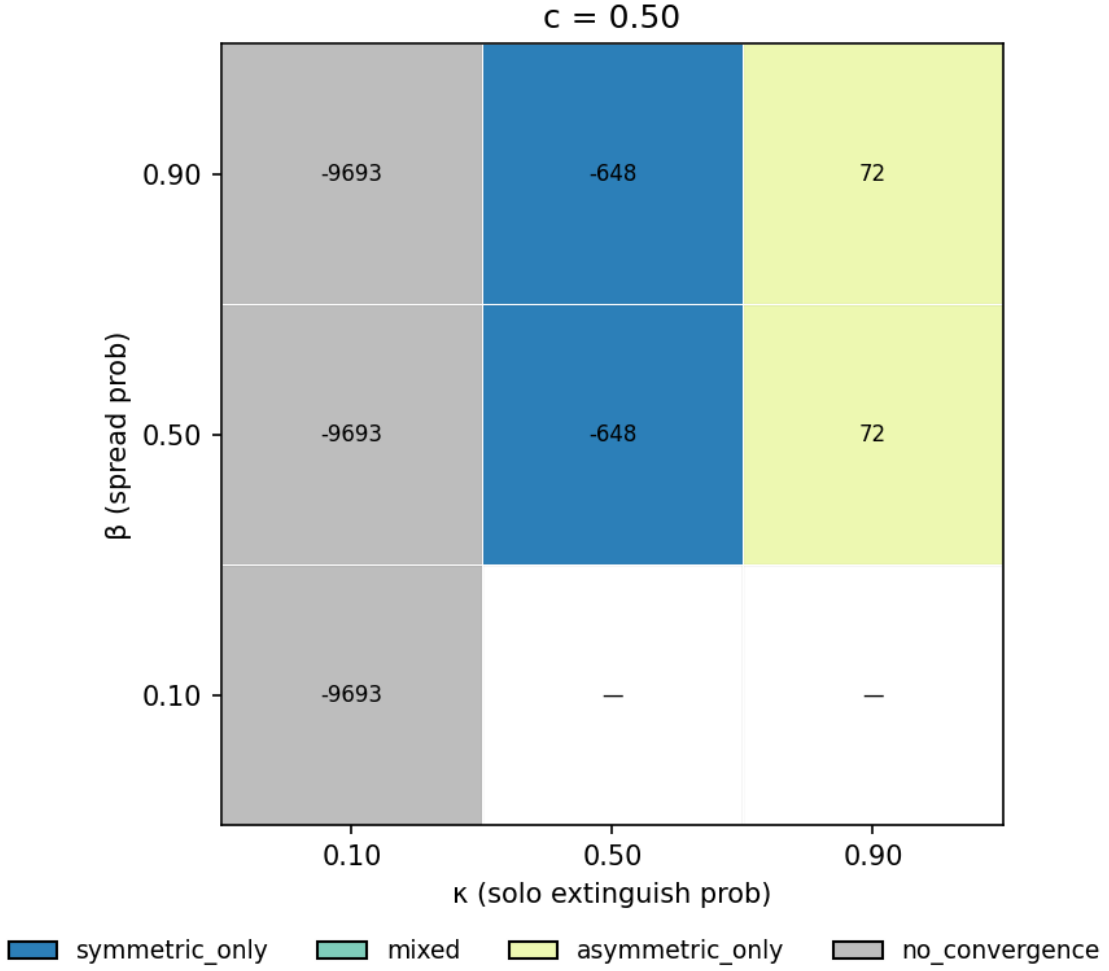
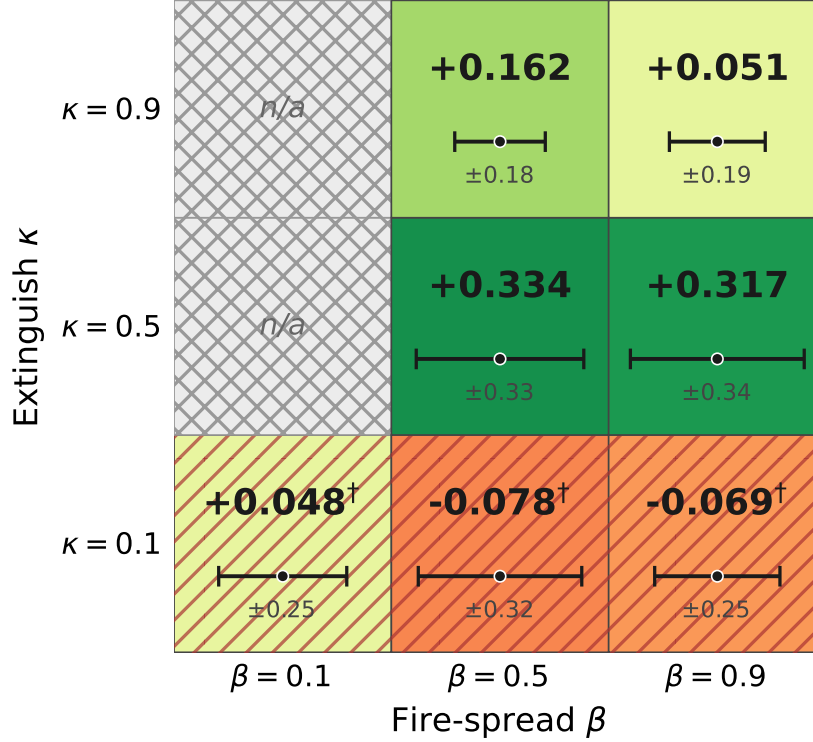


Figure 1: Empirical NE phase diagram for 4-agent Bucket Brigade across (β, κ) at $c=0.5, \rho=0.02, T_{\min}=12$ under the `minimal_specialization` reward tuple. Each cell is the verdict of a heterogeneous Double-Oracle solver over the continuous heuristic-parameter strategy space (Hero, Firefighter, FreeRider, ...). Three regimes appear: `no_convergence` at $\kappa=0.1$ (collapse), `symmetric_only` at $\kappa=0.5$ (Hero NE dominates), and `asymmetric_only` at $\kappa=0.9$ (1 Hero + 3 Firefighters; the rest-trap). The grid is β -independent at $c=0.5$, matching the prediction in (S)–(C) but at substantially different κ thresholds than the closed-form derivation places them.



Horizontal bar in each cell: ± 1 std (4 seeds), scaled so widest std ≈ 0.34 spans 80% of the cell.

- NE-anchored gap_closed_ne
- fallback gap_closed_homogeneous (†, no NE)
- cell not sampled (n/a)

Figure 2: Per-cell PPO `gap_closed_ne` on the 7-cell preview (4 seeds per cell; 50 iterations $\times$ 2048 rollout steps with the in-repo `JointPPOTrainer`). The $\kappa=0.5$ row (symmetric-only NE) shows the largest gap closure (0.326 mean across $\beta \in \{0.5, 0.9\}$), the $\kappa=0.9$ row (asymmetric-only) shows positive but smaller gap closure (0.106), and the $\kappa=0.1$ row (no-pure-NE; hatched cells, † marker) reports `gap_closed_homogeneous` instead because no NE policy exists—values are near zero or negative, confirming PPO fails to recover useful structure. The $\beta=0.1$ column was not sampled in the 7-cell preview. The ordering across rows matches the analytical prediction of §3.